# Digital Image Processing

# Topic

- Working and components inside digital camera
- Pixels
- Image
- Representation
- Sampling
- Quantization

# Digital Camera

- A digital camera captures images by converting light into digital data through a process involving several key components and steps: image acquisition, sampling to create pixels, and quantization to assign color/intensity values

# What is a Pixel in Digital Image Processing?

- Pixel = Picture Element
- It is the smallest unit of a digital image.
- A digital image is made up of thousands or millions of tiny squares called pixels.Each pixel stores numerical values that represent:
- Brightness (Intensity) → for grayscale images
- Color (RGB values) → for color images

# Key Points

- 1. Smallest part of a digital image
- A pixel is the tiny dot that forms the image.
- 2. Contains intensity or color value
- Grayscale pixel → 0 to 255
- Color pixel → 3 values (R, G, B), each 0 to 255
- 3. High number of pixels = High resolution
- More pixels → clearer and sharper image
- Fewer pixels → blurry or blocky image

# Key Points

- 4. Pixel has no physical sizePixel size depends on:
- camera sensor
- display resolution
- 5. Pixel location is represented by (x, y)
- Example: pixel at row 10, column 20 → (10,20

# Pixel in Color Image (RGB)

- Each pixel has 3 components:
- R = Red
- G = Green
- B = Blue
- Example pixel:(255, 120, 60)→ Bright red + little green + little blue

# Pixel in Grayscale Image

- Pixel in Grayscale Image
- Only one value (0–255):
- 0 → Black
- 255 → White
- In-between → Grey levels

# Pixel Depth

- Also called bit depth, it means how many bits are used to store pixel value.
- Examples:
- 8-bit image → 256 intensity levels
- 24-bit color image → 8 bits R + 8 bits G + 8 bits B = 16.7 million colors-

# Note

- A pixel is the smallest picture element of a digital image. It stores intensity or color information as numerical values. Grayscale pixels range from 0–255, while color pixels contain RGB values. Pixels are arranged in rows and columns to form an image. Higher number of pixels results in higher image resolution

- **Image Representation in Dip**

# Image representation

- Image representation in digital image processing is the method of encoding visual information into a digital format, typically by creating a grid of pixels, where each pixel holds information like color or intensity

# Image Representation Types

- 1. Grayscale Image

  One value per pixel

  Range: 0 (black) → 255 (white)
- 2. Binary Image

  Only two values: 0 and 1
- 3. Color Image (RGB)

  3 components per pixel

  Example pixel: (120, 20, 255)
- 4. Multi-spectral / Hyper-spectral Image

  Many spectral bands (satellite imaging)

# Sampling & Quantization

# Sampling

- Sampling in digital image processing is the process of converting a continuous analog image into a discrete image by selecting a finite number of points, or pixels, at regular spatial intervals. It digitizes the spatial coordinates of the image and determines its spatial resolution.

# Sampling

- These are the two steps that convert a continuous image → digital image.
- A Sampling (Spatial Sampling)Sampling = selecting how many pixels will represent the image.
- More sampling → more pixels → high resolution
- Less sampling → blocky image
- Example:1920×1080 image has 2,073,600 samples (pixels)
- Sampling defines image resolution

# Quantization

- Quantization = assigning numeric intensity values to each sample (pixel).
- More quantization levels → smooth image
- Fewer levels → posterization or banding effect
- Common Quantization Levels
- 8-bit grayscale → 256 levels
- 24-bit color → 16.7 million colors